

FORMAL LANGUAGES AND AUTOMATA THEORY [Revised Credit System] (Effective from the academic year 2018-19) SEMESTER - IV			
Subject Code	CSE 2254	IA Marks	50
Number of Lecture Hours/Week	03	Exam Marks	50
Total Number of Lecture Hours	36	Exam Hours	03
CREDITS - 03			
Course objectives: This course will enable students • Understand the mathematical meaning of Grammar • Know how to generate Languages using grammars • Design Automata for various languages			
Module -1			Teaching Hours
INTRODUCTION TO THE THEORY OF COMPUTATION AND FINITE AUTOMATA: Three basic concepts, Some Applications, Deterministic Finite Accepters, Nondeterministic Finite Accepters, Equivalence of Deterministic and Nondeterministic Finite Accepters, Reduction of the Number of States in Finite Automata. Text Book 1: Chapter 1:1.2 - 1.3, Chapter 2: 2.1 - 2.4			08 Hours
Module -2			
REGULAR LANGUAGES, REGULAR GRAMMARS AND PROPERTIES OF REGULAR LANGUAGES: Regular Expressions, Connection between Regular Expressions and Regular Languages, Regular Grammars, Closure Properties of Regular Languages, Identifying Non-regular Languages. Text Book 1: Chapter 3: 3.1 -3.3, Chapter 4: 4.1, 4.3			07 Hours
Module - 3			

CONTEXT-FREE LANGUAGES AND SIMPLIFICATION OF CONTEXT-FREE GRAMMARS AND NORMAL FORMS: Context-Free grammars, Parsing and Ambiguity, Context-Free Grammars and programming languages, Methods for Transforming Grammars, Two important Normal Forms. Text Book 1: Chapter 5: 5.1 -5.3, Chapter 6: 6.1 – 6.2	07 Hours
Module-4	
PUSHDOWN AUTOMATA AND PROPERTIES OF CONTEXT-FREE LANGUAGES: Nondeterministic Pushdown Automata, Pushdown Automata and Context-Free Languages, Deterministic Pushdown Automata and Deterministic Context-Free Languages, Two Pumping Lemmas, Closure properties and Decision Algorithms for Context-Free Languages. Text Book 1: Chapter 7: 7.1 – 7.3, Chapter 8: 8.1 -8.2	06 Hours
Module-5	
TURING MACHINES AND OTHER MODELS OF TURING MACHINES: The Standard Turing Machine, Nondeterministic Turing Machines, Linear Bounded Automata. A HIERARCHY OF FORMAL LANGUAGES & AUTOMATA Recursive and Recursively Enumerable Languages, Unrestricted grammars, Context-Sensitive Grammars and Languages, The Chomsky Hierarchy. Text Book 1: Chapter 9: 9.1, Chapter 10:10.3, 10.5, Chapter 11: 11.1-11.4	08 Hours
Course outcomes:	
After studying this course, students will be able to: 1. Acquire knowledge of fundamental concepts in grammars, formal languages and automata theory 2. Design grammars for different language classes 3. Construct languages from different grammars	

4. Design automata for different language classes
5. Acquire a fundamental understanding of core concepts relating to the theory of computation and computational models

Text Books:

1. Peter Linz, *An Introduction to Formal Languages and Automata*, (6e), Jones & Bartlett Learning, 2016.

Reference Books:

1. John C Martin, *Introduction to Languages and the Theory of Computation*, (3e), McGraw Hill, India, 2007.
2. J E Hopcroft, Rajeev Motwani & Jeffrey D Ullman, *Introduction to Automata Theory, Languages and Computation*, (3e), Pearson Education, 2006.
3. K.L.P. Mishra, N. Chandrashekharan, *Theory of Computer Science*, (3e), PHI publications, 2007.
4. Michael Sipser, *Theory of Computation*, Cengage Learning, 2007.